

Lift Station PdM — Training Data & Setup Review

Pump condition monitoring · setup criteria · proposed PdM extensions

Document ID	MV-PDM-LS-0.1
Status	Implemented — v0.2
Scope	Setup schema, location profiles, lifecycle metadata, PdM extensions, logger popup UI
Seed range	30–360 days via Demo seed days + Seed demo data
Generated	2026-07-25

1. Purpose

This document captures the **design review** from the lift-station PdM training-data discussion. It defines:

- 1. **Setup criteria** MooreVIEW should store for each lift station (pull-down float ladder, simplex / duplex / triplex, location class, pump age, install/repair history).
- 2. **Proposed PdM schema and code changes** — listed for approval; **not implemented** in this phase.
- 3. **180-day synthetic training plan** — deferred until setup schema and PdM extensions are approved.

Review goal: Confirm setup fields, location profiles, and PdM API/storage changes **before** building docs, simulators, or Mongo seed scripts.

2. Summary of prior discussion

MooreVIEW PdM today combines:

Layer	Source	Role
SCADA historian	Mongo <code>pen_sample</code>	Run hours, starts, CT amps, levels
Edge inference	Mongo <code>edge_inference</code>	Per-start signature, anomaly score, label
Feature windows	<code>pdm_features</code> (default 5 min)	Aggregated stats → health index
Forecast	<code>failureForecast.js</code>	RUL from health trend + optional start-time trend

Existing demo path: **Simulate motor start / cap failure** (`motorStartSim.js` , model `single-phase-start-v1`) — 90–180 days of degrading start times.

For **lift-station submersible pumps** (e.g. Pump 2 in a duplex alternator), training data must reflect:

Criterion	Why it matters
Pump age factor	Scales wear rate; same CT signature ≠ same RUL on old vs new pump
Pump starts	Cycle fatigue, electrical inrush events
Pump runtime	Continuous load, seal heat, impeller erosion
Application: lift station	Float ladder, alternator, fail-to-run, FOG/solids
Location class	Load pattern and contaminant prior (mall vs school vs restaurant)

Duplex **lag pump** (Pump 2) has fewer starts but often longer runs when `LVL_LAG` / `LVL_HIGH` demand both pumps — it must be modeled as a **separate PdM asset**, not merged with Pump 1.

3. Lift station physical model (pull-down)

3.1 Float ladder — pump-down (pull-down)

MooreVIEW duplex logic uses cumulative floats (liquid **at or above** float = ON):

Float	Tag (duplex)	Meaning
Off / stop	LVL_OFF	Wet well drained — pumps off
Lead	LVL_LEAD	First demand — lead pump runs
Lag	LVL_LAG	Second demand — lead + lag
High alarm	LVL_HIGH	Critical level — all online pumps + alarm

Pull-down behavior: As level falls, floats drop in reverse order (high → lag → lead → off). Training data should include level-driven **start/stop edges**, not random Poisson starts.

3.2 Pump configurations

Config	Pumps	MooreVIEW program (reference)	Alternator	Typical use
Simplex	1	logic/lift_simplex / single-pump ST	N/A	Small tenant, low flow
Duplex	2	logic/36_duplex_lift_station.st / 37_duplex_lift_station_parcc_st.st	ALT2 lead/lag	Mall, restaurant, ALF
Triplex	3	logic/28_alternator_triplex.st / triplex fixtures	ALT3 / ALT4	School campus, large strip center

Per-pump SCADA tags (duplex example — extend numbering for triplex):

Field	Pump 1	Pump 2	Pump 3 (triplex)
Run hours	MOTOR1_HRS	MOTOR2_HRS	MOTOR3_HRS
Start count	MOTOR1_STARTS	MOTOR2_STARTS	MOTOR3_STARTS
CT phase A	I1_RAW / AI1	I4_RAW / AI4	per I/O map
Run feedback	P1_RUN_FB	P2_RUN_FB	P3_RUN_FB
Alternator role	lead/lag rotate	lead/lag rotate	lead/lag/lag2

4. Location variation profiles

Each site gets a **location class** that drives simulation priors and PdM context metadata. Values are starting points for review — field calibration can adjust.

Class	ID	Peak flow pattern	FOG / solids index (0–1)	Rain sensitivity	Notes
Strip mall	strip_mall	Business hours + dining eve	0.55	High (large roof)	Multiple restaurants, shared interceptor
School	school	Class hours; low summer	0.25	Medium	Seasonal idle; duplex common
Assisted living (ALF)	alf	Steady 24/7; meal peaks	0.40	Medium	Consistent load; reliability critical

Convenience store	convenience_store	Moderate; spikes from restrooms	0.35	Low–medium	Often simplex; grease from food prep
Standalone restaurant	restaurant	Lunch + dinner peaks	0.75	Medium	Highest FOG; ragging/clog risk

Derived simulation parameters (review defaults):

Parameter	strip_mall	school	alf	convenience_store	restaurant
Base starts/day (lead pump)	22	14	18	10	26
Base starts/day (lag pump)	10	6	8	—	12
Avg run length (min)	6–12	8–15	10–18	5–10	4–10
Material load multiplier	1.25	0.85	1.05	1.00	1.45
Weekend factor	0.85	0.15	1.0	1.0	1.1

5. Pump age and lifecycle

5.1 Age factor

Age is metadata + counter initialization, not a live sensor.

$$\text{ageYears} = (\text{referenceDate} - \text{installDate}) / 365.25$$

$$\text{ageFactor} = 1 + 0.04 \times \max(0, \text{ageYears} - 3)$$

Age band	Years	ageFactor (approx)	Training use
New	0–3	1.00	Baseline signatures
Mid-life	4–8	1.04–1.20	Normal wear slopes
Aged	9–14	1.24–1.44	Accelerated degradation
End of life	15+	1.48+	Pre-failure scenarios

Equivalent age (optional): $\text{ageEqHours} = \text{MOTORx_HRS} + 0.02 \times \text{MOTORx_STARTS}$ — useful when install date unknown.

5.2 Installation and repair history

Each pump maintains a **service timeline**. Repairs reset or partially reset specific failure modes but do not always reset age.

Event type	ID	Effect on metadata
Original install	install	Sets <code>installDate</code> ; baseline counters optional
Pump replacement	pump_replacement	New <code>installDate</code> for that pump asset; reset <code>MOTORx_HRS</code> / <code>MOTORx_STARTS</code> in field

Seal repair	seal_service	Resets seal-wear contribution; age factor unchanged
Impeller / wear ring	impeller_service	Resets hydraulic wear; may improve run amps
Motor / starter	motor_or_starter	Resets electrical start signature baseline
Cable / splice	cable_repair	Resets phase imbalance baseline
Clog / jetting	clog_clearing	Operational event; not a hardware refresh

Review rule: PdM context uses **effective install date** = date of last `pump_replacement` , else original `install` .

6. Proposed setup data schema (for review)

Stored per site in project data (proposed file: `data/lift-stations.json` or embedded in project `.est` archive). PdM reads **asset context** at feature-build time.

6.1 Site record

```
{
  "siteId": "ls-strip-mall-01",
  "name": "Westfield Lift Station #1",
  "configuration": "duplex",
  "floatLadder": "pull_down",
  "locationClass": "strip_mall",
  "coordinates": { "lat": 28.05, "lng": -82.45 },
  "commissionedDate": "2024-03-15",
  "notes": "Shared grease interceptor; 4 restaurant tenants",
  "pumps": ["pump-1", "pump-2"]
}
```

6.2 Pump asset record

```
{
  "assetId": "pump-2",
  "siteId": "ls-strip-mall-01",
  "pumpIndex": 2,
  "role": "lag",
  "configuration": "duplex",
  "locationClass": "strip_mall",
  "installDate": "2017-06-01",
  "manufacturer": "Example Submersible Co",
  "model": "4DF-15",
  "hp": 5,
  "scadaTags": {
    "runHours": "MOTOR2_HRS",
    "starts": "MOTOR2_STARTS",
    "runFeedback": "P2_RUN_FB",
    "ctPhaseA": "I4_RAW",
    "ctPhaseB": "I5_RAW",
    "ctPhaseC": "I6_RAW"
  },
  "pdm": {
    "modelId": "lift-submersible-v2",
    "enabled": true
  },
  "serviceHistory": [
    {
      "date": "2017-06-01",
      "type": "install",
      "vendor": "ABC Plumbing",
      "notes": "Original duplex install"
    },
    {
      "date": "2022-11-08",
      "type": "seal_service",
      "vendor": "ABC Plumbing",
      "notes": "Upper seal replaced; wet well cleaned"
    },
    {
      "date": "2025-04-02",
      "type": "clog_clearing",
      "vendor": "Jetting Services",
      "notes": "FOG slug after interceptor bypass"
    }
  ]
}
```

6.3 Computed context (written to edge inference / feature docs)

At build or simulate time:

```
{
  "context": {
    "siteId": "ls-strip-mall-01",
    "assetId": "pump-2",
    "configuration": "duplex",
    "pumpIndex": 2,
    "pumpRole": "lag",
    "locationClass": "strip_mall",
    "application": "lift_station",
    "floatLadder": "pull_down",
    "installDate": "2017-06-01",
    "effectiveInstallDate": "2017-06-01",
    "pumpAgeYears": 8.2,
    "ageFactor": 1.21,
    "lastServiceType": "clog_clearing",
    "lastServiceDate": "2025-04-02",
    "materialProfile": {
      "fogIndex": 0.55,
      "regime": "fog_heavy"
    }
  }
}
```

6.4 Configuration tag map (PdM asset → SCADA)

Simplex

```
{
  "pump-1": ["MOTOR1_HRS", "MOTOR1_STARTS", "I1_RAW", "I2_RAW", "I3_RAW", "LVL_LEAD",
    "LVL_HIGH"]
}
```

Duplex

```
{
  "pump-1": ["MOTOR1_HRS", "MOTOR1_STARTS", "I1_RAW", "I2_RAW", "I3_RAW", "P1_RUN_FB"],
  "pump-2": ["MOTOR2_HRS", "MOTOR2_STARTS", "I4_RAW", "I5_RAW", "I6_RAW", "P2_RUN_FB"]
}
```

Triplex

```
{
  "pump-1": ["MOTOR1_HRS", "MOTOR1_STARTS", "I1_RAW", "I2_RAW", "I3_RAW"],
  "pump-2": ["MOTOR2_HRS", "MOTOR2_STARTS", "I4_RAW", "I5_RAW", "I6_RAW"],
  "pump-3": ["MOTOR3_HRS", "MOTOR3_STARTS", "I7_RAW", "I8_RAW", "I9_RAW"]
}
```

(Exact CT tag names follow project I/O map — Parc ST duplex uses `I1_RAW` – `I6_RAW` .)

7. Proposed PdM changes (review only — not implemented)

7.1 Settings extension (`settings.json` → `pdm`)

Current shape (`pdmSettings.js`):

```
{
  "assetTags": { "motor-202": ["MOTOR_START_MS", "MOTOR_CURRENT"] },
  "windowMin": 5,
  "featuresCollection": "pdm_features",
  "failureThreshold": 0.3,
  "buildEnabled": false,
  "buildIntervalHours": 24
}
```

Proposed additions:

Field	Type	Purpose
<code>assetContext</code>	<code>{ [assetId]: LiftPumpContext }</code>	Static setup: location, config, install date, role
<code>siteRegistryPath</code>	string	Optional path to <code>lift-stations.json</code>
<code>liftModelId</code>	string	Default edge model id (<code>lift-submersible-v2</code>)
<code>forecastMethods</code>	string[]	Enable <code>health_index</code> , <code>run_amps_creep</code> , <code>start_time_ms</code>

7.2 New modules (planned)

Module	Responsibility
<code>src/pdm/liftStationSetup.js</code>	Load/validate site + pump records; compute <code>ageFactor</code> , effective install
<code>src/pdm/liftPumpContext.js</code>	Merge setup + live counters into context blob
<code>src/pdm/liftPumpSim.js</code>	180-day generator: pull-down cycles, location profile, degradation
<code>src/pdm/liftForecast.js</code>	Extend <code>failureForecast</code> for run-amp creep + clog labels

7.3 API routes (planned)

Method	Path	Purpose
GET	/pdm/lift-setup	List sites / pumps
PUT	/pdm/lift-setup	Save site or pump record + service history
POST	/pdm/sim/lift-pump	Seed Mongo with simulated 180-day sample
GET	/pdm/context/:assetId	Resolved context for an asset (review/debug)

7.4 Feature alignment changes

`featureAlign.js` — attach `context` and per-window derived metrics:

- `deltaRunHours` , `startsInWindow` , `avgRunAmps` , `dutyCycle`
- Copy from `assetContext` at build time

7.5 UI (planned, post-approval)

- Historian** → **PdM** → **Lift station setup** tab: site form, pump table, service history editor
- Location class dropdown; configuration simplex/duplex/triplex
- Install date + repair log with type picker
- Simulate lift pump (180d)** button (disabled until schema approved)

7.6 Mongo collections (unchanged names)

Collection	Addition
pen_sample	No schema change
edge_inference	Optional top-level <code>context</code> object
pdm_features	Optional <code>context</code> + derived per window

8. Training data plan (deferred)

After approval of Sections 6–7:

Deliverable	Description
Age-stratified fleet	6 assets × ages {1, 3, 5, 8, 12, 15} yr
Location matrix	5 location classes × duplex lag pump
Duration	180 days, 5-min windows
Labels	RUL, failure mode (<code>impeller_worn</code> , <code>seal_leak</code> , <code>clog_ragging</code> , ...)
Export	PdM CSV + optional ML export JSON

Not started until this review document is signed off.

9. Review checklist

Use this section in the review meeting. Mark ☐ open · ☒ approved · ☐ revise.

9.1 Setup schema

- ☐ Site record fields (`siteId` , `configuration` , `locationClass` , `floatLadder`)
- ☐ Pump record fields (`installDate` , `role` , `scadaTags` , `serviceHistory`)
- ☐ Service event types (`install` , `pump_replacement` , `seal_service` , ...)
- ☐ Effective install date rule after pump replacement
- ☐ Age factor formula ($1 + 0.04 \times \max(0, \text{age} - 3)$)

9.2 Location profiles

- ☐ Five classes: `strip_mall`, `school`, `alf`, `convenience_store`, `restaurant`
- ☐ Default starts/day and FOG index table (Section 4)
- ☐ Material regime labels for edge inference context

9.3 Configuration coverage

- ☐ Simplex tag map
- ☐ Duplex tag map (Pump 1 lead / Pump 2 lag)
- ☐ Triplex tag map
- ☐ Pull-down float ladder reflected in sim (not random starts)

9.4 PdM code scope

- ☐ `assetContext` in `settings.json` vs separate `lift-stations.json`
- ☐ New `/pdm/lift-setup` and `/pdm/sim/lift-pump` routes
- ☐ `lift-submersible-v2` model id and failure mode labels
- ☐ Extended failure forecast (run amps + health index)
- ☐ UI tab in Historian → PdM (phase 2)

9.5 Out of scope confirmation

- ☐ No sample data seeded in this phase
- ☐ No production CMMS rule changes
- ☐ No edge firmware model deployment

10. Sign-off

Role	Name	Date	Notes
Product / domain			

PdM / data			
Engineering			

Next step after approval: Implement `liftStationSetup.js` + schema validation, then `liftPumpSim.js` and first 180-day duplex lag-pump seed for one strip-mall site.

Appendix A — Failure mode labels (proposed)

Label	Typical signature	Location sensitivity
healthy	Stable run amps, normal starts	All
impeller_worn	Run amp creep 10–25% over months	All; faster with grit
seal_leak	Longer starts, thermal drift	Aged pumps
clog_ragging	Short cycles, restart bursts	restaurant, strip_mall
bearing_wear	Current ripple (if modeled)	Aged, high runtime
phase_imbalance	CT spread on I4–I6	cable_repair resets

Appendix B — Reference: existing PdM demo

Item	Value
Sim endpoint	<code>POST /pdm/sim/motor</code>
Model	<code>single-phase-start-v1</code>
Asset	<code>motor-202</code>
Tags	<code>MOTOR_START_MS</code> , <code>MOTOR_CURRENT</code> , <code>LINE_VOLTAGE</code>
Max days	180 (<code>motorStartSim.js</code>)

Lift-station sim will follow the same Mongo insert pattern but use pull-down level logic and `lift-submersible-v2` context.